

**EXTERNAL LONGITUDINAL TUMESCENCE / RIGIDITY / PHYSIOLOGY WEARABLE —
SAMPLING, QUALITY INDICES, PROCESSING, AND OUTPUTS (NON-LIMITING)**

A. Data Acquisition and Sampling (Non-Limiting)

In various embodiments, an external longitudinal wearable configured for all-day and/or overnight use acquires time-series signals representing physiologic state and tissue-adjacent dynamics over extended periods. The wearable may be configured as a C-shaped device or a substantially closed ring and may acquire signals from one or more sensor modalities including, without limitation, temperature, pressure/contact, strain/deformation, magnetic displacement (hall-effect sensor and magnet), optical sensing (PPG), and inertial sensing (gyroscope/IMU). Non-limiting acquisition approaches include: 1. continuous sampling during wear periods; 2. duty-cycled sampling to reduce power while retaining trend fidelity; 3. burst sampling during candidate windows identified by stability conditions; and 4. event-triggered sampling, wherein sampling rates increase upon detection of transitions in pressure distribution, strain signatures, magnetic displacement signatures, optical coupling, and/or thermal trends. In certain embodiments, the wearable time-stamps data on-node, buffers data locally, and performs periodic synchronization to a computing system.

B. Preprocessing and Signal Conditioning (Non-Limiting)

In various embodiments, a computing system performs preprocessing including: 1. noise reduction filtering applied per modality; 2. baseline normalization and drift correction, including compensation for temperature-dependent sensor behavior; 3. coupling compensation using contact integrity and fit stability estimates; and 4. fusion of multiple modalities to improve robustness of derived metrics. In certain embodiments, preprocessing occurs on-node; in other embodiments it occurs on a user device and/or a remote computing environment.

C. Quality Indices and Validity Gating (Non-Limiting)

In various embodiments, the computing system computes one or more quality indices (QI) to accept, down-weight, or exclude data windows. Non-limiting QI components include: 1. contact integrity index, derived from pressure/contact stability, distribution consistency, and coupling continuity; 2. motion gating index, derived from IMU/gyro magnitude, jerk, and orientation stability; 3. magnetic displacement plausibility, derived from hall-effect and magnet signals indicating consistent geometry/displacement behavior; 4. optical coupling index, derived from PPG signal quality and pulse feature plausibility; 5. thermal plausibility index, derived from temperature stability and on-body coupling dynamics; and 6. signal continuity index, derived from dropout detection, saturation detection, and clipping detection. In certain embodiments, the system defines valid windows as those meeting minimum thresholds across at least two of the above indices and applies higher thresholds for higher-confidence outputs.

D. Artifact Rejection and Robustness (Non-Limiting)

In various embodiments, artifact rejection includes: 1. motion artifact suppression using IMU-informed filtering and window rejection; 2. contact artifact detection using abrupt pressure distribution changes and coupling discontinuities; 3. thermal artifact detection using discontinuity checks and compensation models; and 4. outlier detection relative to within-user baselines. When artifacts are detected, the system may label windows as reduced confidence, exclude them from reporting, and/or request additional valid

measurement windows.

E. Physiologic Confirmation and Anti-Spoof Measures (Non-Limiting)

In various embodiments, the system uses PPG-derived pulse features as a physiologic confirmation signal when available. In certain embodiments, anti-spoof confidence is improved by requiring coherence across modalities including, without limitation: 1. agreement between pressure/strain signatures and magnetic displacement signatures; 2. agreement between coupling dynamics (pressure/contact) and thermal behavior; and/or 3. agreement between detected pulse features and plausible on-body coupling and motion stability. In certain embodiments, outputs intended for correlation (cross-device and/or cross-user, if enabled) are gated such that only windows meeting enhanced anti-spoof thresholds are eligible.

F. Privacy-Preserving Outputs (Node 1) (Non-Limiting)

In various embodiments, the external longitudinal wearable generates privacy-preserving outputs presented without explicit anatomical imagery and without explicit anatomical labeling. Non-limiting output families include: 1. rigidity index derived from pressure/strain/displacement behaviors within validated windows; 2. tumescence pattern index derived from circumferential change signatures and stability scoring; 3. duration and persistence indices derived from validated contiguous windows meeting quality thresholds; 4. stability/variability indices reflecting within-window variation and cross-window repeatability; 5. thermal context tags and temperature-compensated indices; 6. confidence indicators derived from quality indices and validity gating; and 7. trend curves and distributions representing change-over-time across days, nights, or longer intervals.

G. Standalone Use and Optional Integration Hooks (Non-Limiting)

In standalone embodiments, Node 1 operates as an independent product that produces longitudinal outputs, confidence indicators, and trend summaries for a single user without requiring other nodes. In optional multi-node embodiments, derived outputs from Node 1 may be time-aligned and correlated with other nodes under quality gating and consent conditions.